

# Video+

## MMM — Marketing Mix Model

Strategic Insights for the CMO

Audience: CMO & Marketing Science Manager

Model: Hill Saturation Regression + HAC Robust SE | 104 weeks |  $R^2 = 0.9922$

# What the MMM model reveals for Video+

8 dimensions of media intelligence — from saturation to ROI by segment

01

## Channel decomposition

What actually drives subscribers

02

## Seasonality & events

Championships, Cup, peak dates

03

## Saturation curves

Where marginal return drops

04

## Adstock & carryover

The effect that persists after the campaign

05

## Retention vs. Acquisition

Differentiated churn by package

06

## Calibration & significance

HAC SE — real statistical confidence

07

## Budget planning with uncertainty

95% CI per channel for planning

08

## ROI by segment & bundle

Where each dollar works harder

# MMM v2 Scorecard — Fit Quality and Validation

Hill Saturation + HAC Robust SE · Training: 80 weeks · Holdout: 24 weeks

0.9922

$R^2$  train (104 wks)

0.9742

$R^2$  out-of-sample

1.14%

MAPE holdout

430.8

RMSE (subs/week)

## What this means for the CMO

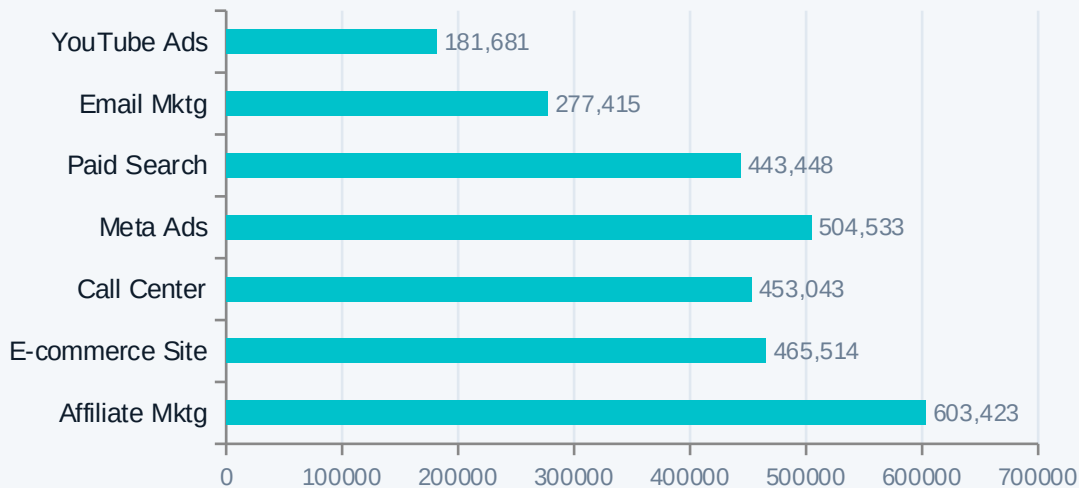
- $R^2 = 0.9922$  in training: the model explains 99% of the weekly variation in new subscribers
- $R^2 = 0.9742$  in holdout (24 unseen weeks): the model generalizes — it does not merely memorize noise
- MAPE = 1.14%: average error of ~315 subscribers per week in out-of-sample forecasts
- All 7 media channels statistically significant at 5% (HAC/Newey-West)

Hill Response =  $x^y / (x^y + \kappa^y)$  · Adstock(t) = spend(t) +  $\delta \times$  Adstock(t-1) · Standard errors: HAC / Newey-West (4 lags)

# What drives subscribers — and what each channel costs

Implicit CPA (MMM) vs. Assumed CPA — deviations reveal where budget is miscalibrated

Subscribers attributed by the model



CPA: MMM vs. Planned

|                 |       |      |
|-----------------|-------|------|
| Affiliate       | R\$22 | -14% |
| E-commerce Site | R\$28 | -60% |
| Call Center     | R\$19 | -53% |
| Meta Ads        | R\$57 | -19% |
| Paid Search     | R\$88 | +97% |
| Email           | R\$19 | -38% |
| YouTube Ads     | R\$54 | -23% |

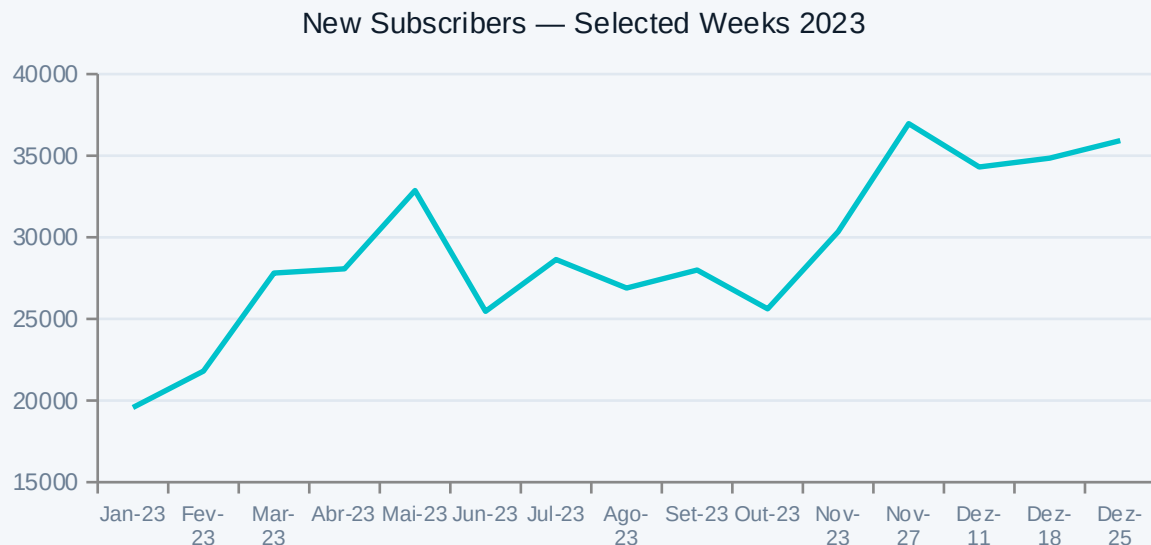


Insight for the CMO: Paid Search received the largest budget (R\$39.2M) but ranked 5th in subscribers. Affiliate Marketing generated 36% more subscribers with 67% less spend.

↑ vs. planned | red = actual CPA higher | green = actual CPA lower

# Sporting events drive +42% spikes in subscriptions

The model isolates seasonal effect from paid media — preventing misattribution



**Week Nov-27/23**

**36.961**

Brazilian Championship Final  
Seasonality Index = 0.6273  
Highest peak of the period

**Dec/23 (3 weeks)**

**~35k/sem**

Seasonal index 0.43–0.45  
Christmas + New Years Eve  
+34% vs. annual average

**Seasonal coefficient**

**+2.825**

new subscribers per  
unit of seasonal index  
( $p < 0.001$ )

💡 Without the model: the CMO would attribute the Nov-27 spike to Google Ads that was active that week. The MMM isolates it: 2,825 subscribers/week come exclusively from the seasonal effect — before any paid media.

# Saturation by channel: where to cut and where to invest more

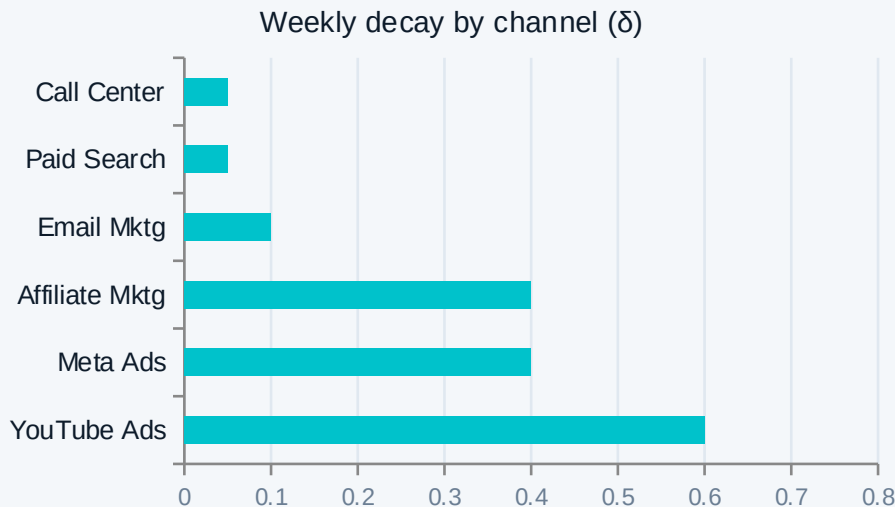
Hill Response =  $x^\delta / (x^\delta + \kappa^\delta) \cdot \gamma$  alto = satura rápido ·  $\kappa$  alto = ponto de saturação mais distante

| Channel              | Sat. Risk | $\gamma$ (curva) | $\kappa$ (meia-sat.) | $\delta$ (decay) | CMO Recommendation                                                     |
|----------------------|-----------|------------------|----------------------|------------------|------------------------------------------------------------------------|
| Paid Search (Google) | HIGH      | 3.0              | 0.8                  | 0.05             | Reduce budget — likely saturated. Marginal return is falling.          |
| E-commerce Site      | HIGH      | 3.0              | 0.4                  | 0.10             | Lowest kappa in the portfolio — saturates early, review spend ceiling. |
| Meta Ads             | MED       | 1.5              | 0.8                  | 0.40             | Maintain cautiously — return still exists but starting to decelerate.  |
| YouTube Ads          | LOW       | 1.0              | 1.6                  | 0.60             | Underinvested — curve still linear, scale with confidence.             |
| Affiliate Marketing  | LOW       | 1.0              | 1.6                  | 0.40             | Best CPA (R\$21) and low saturation — top scaling priority.            |
| Email Marketing      | LOW       | 1.0              | 1.6                  | 0.10             | Low cost, no visible saturation — automation opportunity.              |

# Up to 60% of a campaigns effect persists for weeks

Pausing YouTube or Meta in a low-conversion week is a measurable mistake

**$\text{Adstock}(t) = \text{Spend}(t) + \delta \times \text{Adstock}(t - 1)$  where  $\delta$  is the weekly decay per channel**



## YouTube Ads ( $\delta=0.60$ )

Week 1: 100% of effect

Week 2: 60%

Week 3: 36%

Week 4: 22%

Do not pause in low-conversion weeks

## Meta Ads ( $\delta=0.40$ )

Week 1: 100%

Week 2: 40%

Week 3: 16%

Maintain continuous flow; pauses cost carryover

## Paid Search ( $\delta=0.05$ )

Effect ~100%  
in the activation week

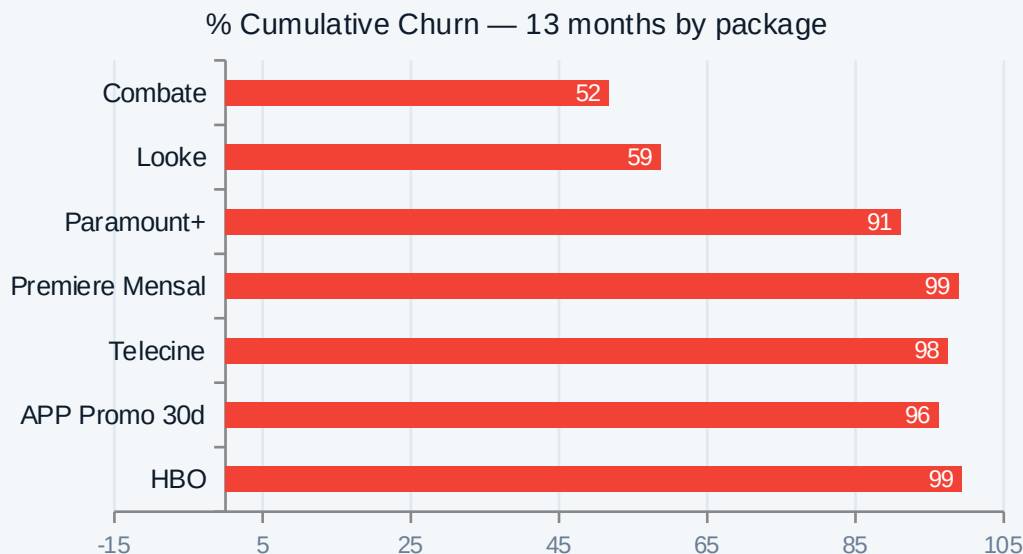
Can be paused without significant  
deferred loss



Rule of thumb: channels with  $\delta > 0.3$  should never be abruptly paused. The lost deferred effect does not show up in the weekly dashboard — but the model measures it.

## 13-month churn: from 51% (Combate) to 99% (HBO)

Investing media in high-churn packages destroys LTV — the model separates each effect



### ● HBO — 99.4% churn

Base praticamente zerada em 13 meses.  
Investimento de aquisição tem retorno quase nulo em LTV.

### ● Premiere Mensal — 98.9%

Assinante de futebol é sazonal.  
Mídia deve focar em retenção na entressafra, não em aquisição.

### ● Combate — 51.8% churn

Base mais fiel do portfólio.  
Alto LTV. Channel prioritário para budget de aquisição.

💡 Golden rule: CPA of R\$167 to acquire an HBO subscriber (LTV ~13 months) vs. R\$118 for Combate (LTV ~10 stable months). The MMM allows calculating churn-adjusted ROI — not just gross CPA.



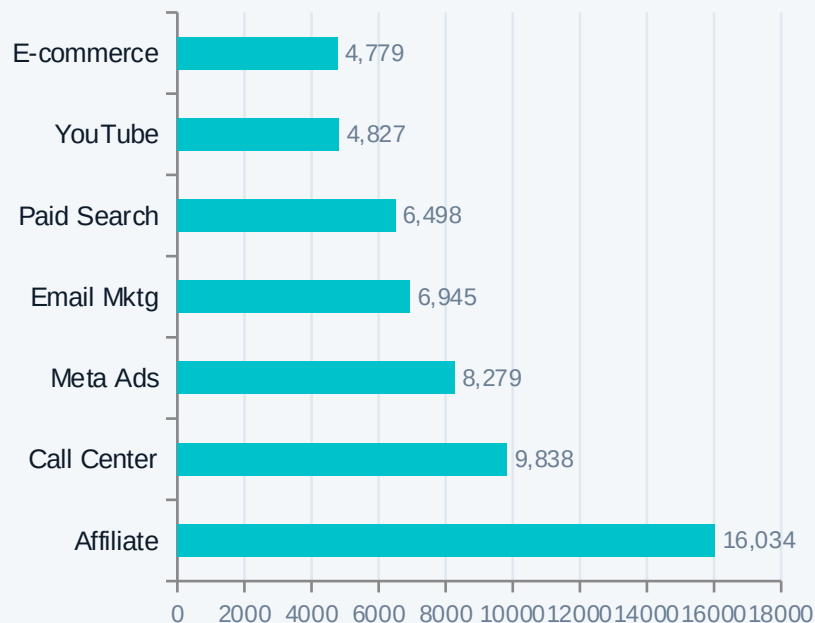
# HAC/Newey-West: real statistical confidence for time series

Durbin-Watson = 1.302 → autocorrelation present → OLS standard errors would be overly optimistic

## VIF — Multicollinearity by channel

|             |      |                                     |
|-------------|------|-------------------------------------|
| Paid Search | 1.78 | ✓ OK (< 5)                          |
| Meta Ads    | 1.57 | ✓ OK (< 5)                          |
| Email Mktg  | 1.64 | ✓ OK (< 5)                          |
| E-commerce  | 1.54 | ✓ OK (< 5)                          |
| Affiliate   | 1.40 | ✓ OK (< 5)                          |
| YouTube     | 1.33 | ✓ OK (< 5)                          |
| Call Center | 1.22 | ✓ OK (< 5)                          |
| Seasonality | 2.19 | ✓ OK (< 5)                          |
| Trend       | 6.38 | ⚠ High — consider orthogonalization |

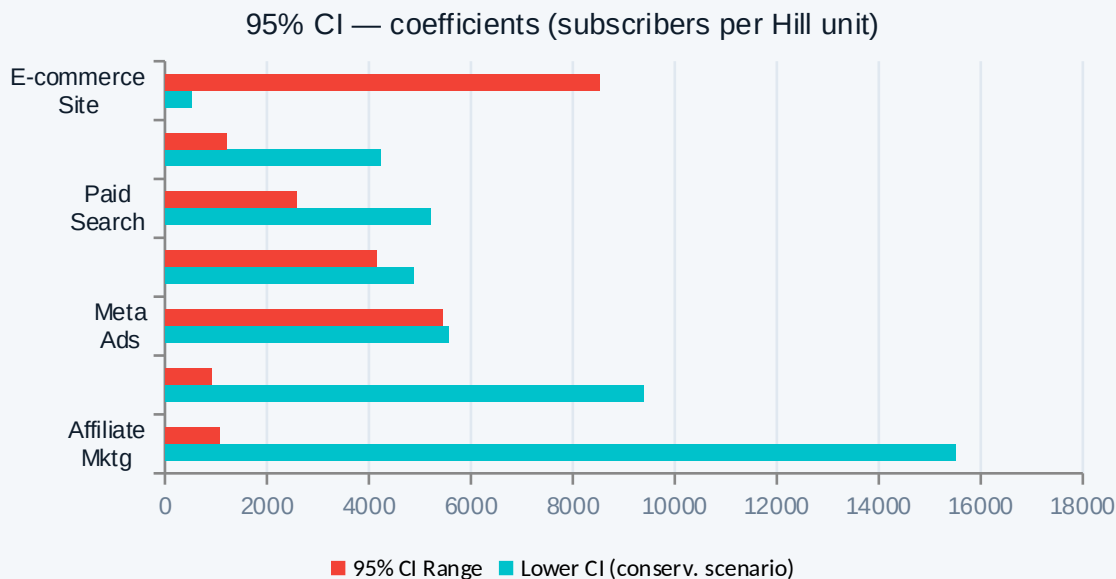
Coefficient by channel (subs per Hill unit)



💡 Affiliate Marketing: coefficient 16,034 with 95% CI = [15,502 – 16,565] — the most precise channel in the portfolio. Meta Ads: CI ±2,720 — plan with a safety margin in budget.

## 95% CI by channel — certainty vs. estimate per channel

Annual planning should use the lower CI as the conservative scenario, not the central coefficient



### Decision guide by channel

HIGH **Affiliate Mktg**

CI  $\pm$ :  $\pm 530$  | Plan without extra margin

HIGH **Call Center**

CI  $\pm$ :  $\pm 460$  | Reliable forecast

MED **Meta Ads**

CI  $\pm$ :  $\pm 2,720$  | Reserve +20% in budget

MED **Email Mktg**

CI  $\pm$ :  $\pm 2,070$  | Reserve +15% in budget

LOW **E-commerce**

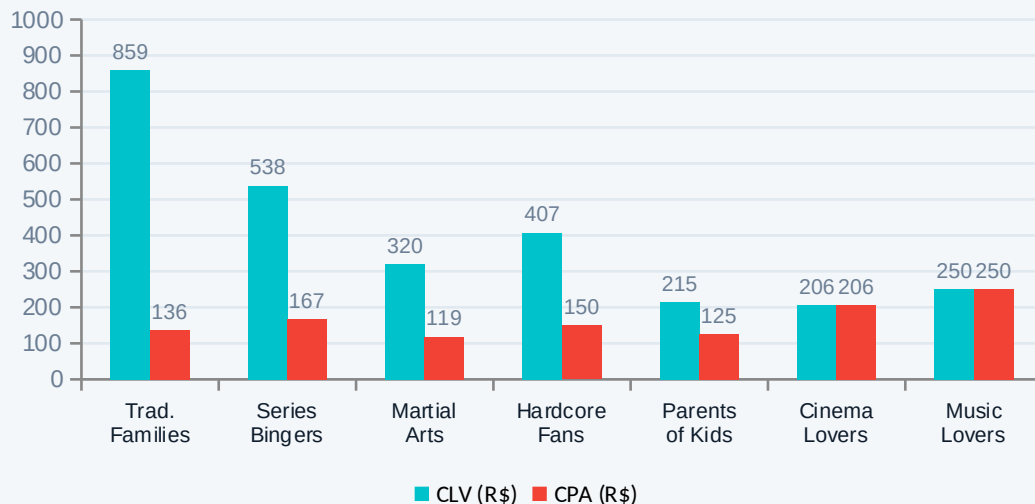
CI  $\pm$ :  $\pm 4,260$  | Reserve +35% — high variance

💡 price\_increase\_flag: coef = -285 subs/week (p=0.068) — signal of negative impact, but statistically inconclusive. Not sufficient to drive a pricing decision alone — requires additional testing.

# Trad. Families: ROAS 6.3x • Music & Cinema: negative ROI

The MMM determines which segments create value — and which destroy budget

CLV vs. CPA by segment (R\$)



ROAS by segment

|                 |       |
|-----------------|-------|
| Trad. Families  | 6.31x |
| Series Bingers  | 3.21x |
| Martial Arts    | 2.70x |
| Hardcore Fans   | 2.72x |
| Parents of Kids | 1.72x |
| Cinema Lovers   | 0.59x |
| Music Lovers    | 0.90x |

💡 Recommendation: concentrate 25% of budget on Traditional Families (ROAS 6.3x) and 20% on Martial Arts (lowest CPA: R\$119). Eliminate or reduce to 5% the budget for Cinema and Music Lovers — both with negative ROI.

# Recommendations for the Video+ CMO

Summary of the 8 MMM dimensions — priority actions

## Rebalance media budget

Reduce Paid Search (saturated,  $\gamma=3$ ) · Increase Affiliate (+36% subscribers with -67% spend)

## Plan around the sports calendar

Championship generates +42% subscribers/week · Concentrate media in the 4 final weeks

## Keep high-adstock channels active

YouTube ( $\delta=0.6$ ) and Meta ( $\delta=0.4$ ) should not be paused · Carryover loss is invisible in the dashboard

## Prioritize high-LTV segments

Traditional Families (ROAS 6.3x) and Martial Arts (CPA R\$118) · Cut Music and Cinema (negative ROI)

## Use 95% CI in planning

Affiliate (CI  $\pm 530$ ): plan without margin · E-commerce (CI  $\pm 4,260$ ): reserve +35% in budget

## Monitor VIF of Trend (6.38)

Orthogonalize trend vs. control variables · Next model version should address this

# Video+

## Marketing Mix Model

### Next steps

- Replace synthetic data with real weekly spend by channel
- Implement Bayesian hierarchical framework (PyMC-Marketing / Robyn)
- Calibrate with geo-experiments (incrementality tests)
- Orthogonalize Trend variable (VIF = 6.38) in model v3
- Run budget scenarios for Q1/Q2 with 95% CI per channel

$R^2 = 0.9922$  · MAPE holdout = 1.14% · 7 significant channels